Synopsis

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Smart Dustbin

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by

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Certificate of Approval

This is to certify that Shaur Shaikh , Suyash Hatankar , Kanishk Patil , Siddhi Aghav are bonafide students of B. Tech in Computer Science and Engineering(Data Science) - Department, St. John College of Engineering and Management, Mumbai. They have satisfactorily completed the requirements of Synopsis as prescribed by ST. JOHN COLLEGE OF ENGINEERING AND MANAGEMENT (An Autonomous College affiliated to University of Mumbai), while working on "Smart Dustbin". They have prepared this synopsis on "Smart Dustbin (Hardware)" in partial fulfilment of the academic requirements.

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ABSTRACT

The Smart Dustbin is an automated, touch-free waste-disposal system designed to improve convenience and reduce direct contact with the lid of a conventional waste bin. The system is built around an Arduino Uno microcontroller, an HC-SR04 ultrasonic distance sensor and a servo motor. The sensor detects a nearby hand or object by transmitting ultrasonic waves and measuring the time taken for the reflected echo to return. The Arduino converts the echo duration into an approximate distance and compares the result with a programmed threshold of 20 centimetres.

When an object is detected within the threshold, the Arduino sends a command to the servo motor to rotate to 90 degrees, opening the lid. When the object is outside the threshold, the servo is commanded to return to 0 degrees and close the lid. The prototype uses a 9 V battery and battery connector, jumper wires, and a bin with a lid connected mechanically to the servo. The program also displays the calculated distance on the Serial Monitor at 9600 baud, which helps the user observe sensor behaviour and troubleshoot the circuit.

The project demonstrates the integration of sensing, embedded programming and mechanical actuation in an accessible prototype. It is suitable for learning basic automation, digital input/output, distance calculation and servo control. The current version focuses on automatic lid operation only; it does not include a bin-full sensor, buzzer, Wi-Fi notifications or waste segregation. The reliability of the prototype depends on proper wiring, suitable power delivery, sensor placement, the weight and movement of the lid, and repeated testing.

CHAPTER 1: INTRODUCTION

1.1 Background

Waste disposal is a routine activity in homes, schools, colleges, offices, laboratories and public spaces. A dustbin provides a place to collect waste before it is transferred for disposal or processing. In a conventional bin, the user may need to lift a lid or press a pedal. Some bins have no lid at all. Although these designs are simple, they can be inconvenient when a person is carrying objects, has occupied hands, or wants to avoid touching a surface that may have accumulated dirt.

Automation offers a way to make common activities more convenient by using sensors and controllers to perform a task with limited human intervention. A Smart Dustbin is an example of a small automated system. Instead of asking the user to operate the lid manually, the bin detects a nearby object and opens its lid automatically. After the object moves away, the system returns the lid to the closed position according to the program.

The project uses an HC-SR04 ultrasonic sensor for proximity detection. The sensor emits high-frequency sound waves and measures the time taken for the echo to return from an object. An Arduino Uno acts as the central controller. It processes the measured duration, calculates the approximate distance and compares that value with a preset threshold. A servo motor converts the controller's signal into mechanical movement and operates the lid.

1.2 Problem Statement

Manual dustbin lids require direct operation and may become inconvenient in situations where hands are occupied or clean handling is preferred. Repeated touching can transfer dirt from the lid to the hands, while a lid that is not closed properly may expose the waste. The aim of this project is to construct a low-cost prototype that detects an approaching hand or object and operates the lid automatically, reducing the need for direct contact.

1.3 Aim of the Project

To design and implement a touch-free Smart Dustbin using an Arduino Uno, HC-SR04 ultrasonic sensor and servo motor, with automatic lid operation based on a distance threshold of 20 cm.

1.4 Objectives

- To build a working prototype using readily available electronic and mechanical components.

- To detect the presence of a nearby hand or object without requiring physical contact with the dustbin.
- To measure object distance using the HC-SR04 ultrasonic sensor.
- To program the Arduino Uno to compare the measured distance with a defined threshold.
- To control the opening and closing movement of the lid using a servo motor.
- To display distance measurements on the Serial Monitor to support observation and troubleshooting.
- To understand the relationship between sensor input, program logic and actuator output.
- To identify possible improvements that could make the system more reliable and useful.

1.5 Need and Significance

A touch-free lid can make waste disposal more convenient and can reduce the frequency with which users need to touch the bin lid. The project is also useful from an educational perspective because it combines several fundamental electronics concepts in a single application. Students can observe how a physical event is converted into a sensor reading, how a microcontroller evaluates that reading and how a motor responds to the resulting command.

The use of a distance threshold makes the system's behaviour easy to understand and adjust. The threshold can be modified in the program to suit the physical arrangement of the bin. The Serial Monitor provides feedback during testing, making it easier to determine whether the sensor is detecting objects and whether the program is responding as intended.

1.6 Scope of the Project

The present prototype focuses on detecting an object within approximately 20 cm and opening or closing the lid. It includes an Arduino Uno, HC-SR04 sensor, servo motor, battery connector, 9 V battery, jumper wires and a bin. The scope includes hardware assembly, programming, basic functional testing and observation of sensor readings. The design does not currently include a system to measure the quantity of waste, send phone notifications, classify waste, or alert the user when the bin is full.

1.7 Applications

- Homes and kitchens where touch-free lid operation may improve convenience.
- Classrooms, college laboratories and offices for demonstrating small-scale automation.

- Educational exhibitions and project demonstrations involving sensors and microcontrollers.
- Prototype development for exploring future smart waste-management features.

1.8 Limitations of the Present Scope

The prototype is not a complete municipal or industrial waste-management system. It does not provide data on bin capacity or collection schedules. Its operation depends on sensor readings and mechanical movement, and performance may vary with the object, sensor position, lid weight, power supply and wiring quality. Additional engineering and testing would be required before adapting the design for heavy-duty or public use.

CHAPTER 2: HARDWARE COMPONENTS AND WORKING PRINCIPLE

2.1 Overview of the Hardware

The Smart Dustbin consists of a sensing section, a control section, an actuation section and a physical bin structure. The HC-SR04 sensor forms the sensing section by measuring the distance to an object. The Arduino Uno forms the control section and executes the decision-making program. The servo motor forms the actuation section by moving the lid. The battery and connector supply power, while jumper wires provide the electrical connections between the modules.

2.2 Arduino Uno

The Arduino Uno is a microcontroller development board used to read inputs and control outputs according to a program. In this project, it receives the echo signal from the HC-SR04 and uses the duration of that signal to estimate distance. It then compares the distance with the programmed maximum range. Depending on the result, it sends a control signal to the servo motor. The Arduino also sends the distance value to a computer through serial communication.

The program uses digital pin 9 as the sensor trigger output, digital pin 10 as the echo input and digital pin 11 as the servo signal pin. The setup() function configures these connections when the board starts. The loop() function then repeats the sensing and control sequence continuously.

2.3 HC-SR04 Ultrasonic Sensor

The HC-SR04 is an ultrasonic ranging module commonly used to estimate the distance of nearby objects. It has power, ground, trigger and echo connections. To start a measurement, the controller sends a short trigger pulse. The module emits a burst of ultrasonic sound waves, which travel through the air and reflect from a nearby object. The returning sound is detected by the sensor, which produces an echo pulse whose duration is related to the distance.

The sensor does not need to physically touch the object. The Arduino measures the duration of the echo pulse and converts it into distance. In the supplied program, the calculation is written as distance = (duration / 2) / 29.1. The factor of two accounts for the outgoing and returning travel of the sound wave. The result is an approximate distance in centimetres, not a direct measurement of the object's size or identity.

2.4 Servo Motor

A servo motor is an actuator that can move its shaft to a commanded angular position. In the Smart Dustbin, the servo is attached to the lid through a mechanical linkage. The program uses servo.write(90) for the open position and servo.write(0) for the closed position. These values are starting points; the actual angles may need to be changed according to the servo mounting, hinge location and lid movement.

The lid should be light enough for the selected servo and should move freely. If the lid is heavy, the linkage is poorly aligned or the hinge creates excessive friction, the servo may struggle or fail to reach the intended position. The motor should be fixed securely, and its moving parts should not trap the wires.

2.5 Battery Connector and 9 V Battery

The battery connector allows the 9 V battery to be connected to the circuit conveniently. A battery provides portable power for a demonstration, but the supply arrangement must be chosen carefully. The Arduino, ultrasonic sensor and servo have specific voltage and current requirements. In particular, a servo can draw a relatively high current when it starts moving or encounters resistance. The power source must therefore be suitable for the connected loads.

The circuit should be wired with correct polarity and a common ground between the controller, sensor and servo signal reference. Before operating the prototype, inspect the connector and wires for loose contacts. If the Arduino resets or the servo behaves unpredictably during movement, the power supply and current demand should be checked.

2.6 Jumper Wires

Jumper wires connect the sensor's trigger and echo pins, the servo signal pin and the power and ground connections. Clear wiring makes the circuit easier to inspect and troubleshoot. Wires should be kept away from the servo's moving arm and the lid hinge. Loose connections may cause intermittent sensor readings or unexpected motor behaviour.

2.7 Bin and Lid Mechanism

The bin is the physical structure that holds waste. Its lid is linked to the servo motor so that angular movement of the servo opens or closes the lid. The mounting arrangement should allow the lid to move smoothly. The sensor should be placed so it can detect a hand approaching the bin without being blocked by the lid itself. The electronics should be secured away from waste and moisture, because exposure can damage the components.

2.8 Pin Connections

- HC-SR04 TRIG pin — Arduino digital pin 9.

- HC-SR04 ECHO pin — Arduino digital pin 10.
- Servo signal wire — Arduino digital pin 11.
- Sensor power and ground — connect to the appropriate supply and ground terminals.
- Servo power and ground — connect according to the servo's requirements; maintain a common ground with the Arduino.

2.9 Working Principle: Step-by-Step

Step 1: When power is applied, the Arduino begins running the program. Serial communication is initialized at 9600 baud, the trigger pin is set as an output, the echo pin as an input and the servo is attached to pin 11.

Step 2: The Arduino sends a trigger pulse to the HC-SR04. The sensor emits ultrasonic waves that travel towards nearby objects.

Step 3: When the sound reflects from an object and returns, the sensor produces an echo pulse. The Arduino uses pulseIn() to measure the time for which the echo signal remains HIGH.

Step 4: The program converts the echo duration to an approximate distance in centimetres using the formula included in the code.

Step 5: The calculated distance is compared with MAX_DISTANCE, which is set to 20 cm. If the distance is less than or equal to 20 cm, the servo is commanded to 90 degrees and the lid opens.

Step 6: If the measured distance is greater than the threshold, the servo is commanded to 0 degrees and the lid closes. The program prints the distance value to the Serial Monitor, waits for one second and repeats the measurement cycle.

2.10 System Flow

Power on → initialize Arduino, sensor and servo → send trigger pulse → receive echo → calculate distance → compare distance with 20 cm → if within range, open lid → otherwise, close lid → display distance → wait briefly → repeat.

CHAPTER 3: SOFTWARE DESIGN, PROGRAM AND TESTING

3.1 Software Overview

The control logic is implemented in an Arduino sketch. The program includes the Servo library, defines the three signal pins and sets the maximum detection distance to 20 cm. The same basic sequence is repeated in the main loop: trigger the ultrasonic sensor, measure the echo, calculate the distance, choose the servo position and display the measured value.

3.2 Libraries and Definitions

The statement #include <Servo.h> makes the Servo library available so that the program can control the motor using simple commands. TRIGGER_PIN is defined as 9, ECHO_PIN as 10, SERVO_PIN as 11 and MAX_DISTANCE as 20. Using named constants makes the sketch easier to read and allows the pin assignments or detection threshold to be changed in one place.

3.3 setup() Function

The setup() function runs once after the Arduino is powered or reset. Serial.begin(9600) enables communication with the Serial Monitor. pinMode(TRIGGER_PIN, OUTPUT) configures the trigger pin to send a pulse, while pinMode(ECHO_PIN, INPUT) configures the echo pin to receive the sensor signal. servo.attach(SERVO_PIN) associates the servo object with the selected output pin.

3.4 loop() Function

The loop() function repeats for as long as the Arduino is running. The trigger pin is first set LOW for two microseconds, then set HIGH for ten microseconds and returned LOW. This sequence asks the HC-SR04 to start a distance measurement. The pulseIn(ECHO_PIN, HIGH) statement measures the duration of the echo pulse and stores it in the duration variable.

3.5 Distance Calculation

The program calculates distance using distance = (duration / 2) / 29.1. The measured duration represents the travel time of the sound pulse to the object and back. Dividing by two estimates the one-way travel time, and the constant 29.1 is used by the supplied code to convert the duration to centimetres. The result is stored in the distance variable and used in the conditional statement.

3.6 Decision and Servo Control

The statement if (distance <= MAX_DISTANCE) checks whether the detected object is within the threshold. If the condition is true, servo.write(90) commands the servo to the open position and delay(1000) pauses for one second. If the condition is false, servo.write(0) commands the servo to the closed position. These angular values can be calibrated after the servo is physically attached to the lid.

3.7 Serial Monitor Output

The program prints the label 'Distance:', the calculated number and the unit 'cm' using Serial.print() and Serial.println(). The Serial Monitor must be set to 9600 baud to display the output correctly. The values help the builder confirm that the sensor is detecting changes as an object approaches or moves away. If the output remains constant or shows implausible values, the sensor wiring, object placement and power connections should be checked.

3.8 Testing Procedure

- Check that the Arduino, sensor, servo, battery connector and jumper wires are connected correctly.
- Inspect the servo linkage and move the lid gently by hand with power disconnected to confirm that it is not jammed.
- Upload the sketch to the Arduino Uno.
- Open the Serial Monitor and select 9600 baud.
- Place an object approximately 10 cm from the sensor and observe the measured distance and lid movement.
- Move the object to approximately 20 cm and then beyond 20 cm to observe the programmed threshold behaviour.
- Repeat the test from different positions and check whether the sensor is blocked by the lid or surrounding structure.
- Adjust the servo angles or sensor placement if the lid does not open or close smoothly.

3.9 Troubleshooting

If the lid does not move, confirm the servo signal connection to pin 11, check the program upload and verify that the servo has a suitable power supply. If the sensor does not show reasonable readings, check the trigger and echo connections on pins 9 and 10 and verify the Serial Monitor baud rate. If the servo jitters or the Arduino resets, check for poor contacts, excessive mechanical load or insufficient current from the power source. If the

lid opens at an unexpected distance, inspect the sensor direction and verify that MAX_DISTANCE is set to the intended value.

3.10 Reliability and Safety Considerations

The provided sketch does not set a timeout value for pulseIn(). If no echo is received, the duration may be zero, which can produce an invalid distance estimate. A refined version could check for a missing echo and keep the lid in a safe state until a valid measurement is available. The sensor and Arduino should be protected from moisture and waste, and wires should be routed so they cannot be caught by the lid. Power should be disconnected before changing the wiring.

CHAPTER 4: RESULTS, ADVANTAGES, LIMITATIONS AND FUTURE SCOPE

4.1 Expected Results

When the circuit is assembled correctly and the program is uploaded, the HC-SR04 should measure the distance to a nearby hand or object. If the measured distance is 20 cm or less, the Arduino should command the servo to rotate to 90 degrees and open the lid. If the distance is greater than 20 cm, the servo should be commanded to 0 degrees and close the lid. The distance value should be printed on the Serial Monitor, making it possible to observe how the sensor responds as the object changes position.

The expected demonstration is a repeated cycle of sensing, calculation and actuation. The quality of the result depends on the mounting of the sensor, the alignment of the servo, the lid's weight and friction, and the stability of the power supply. Actual performance should be recorded during testing rather than assumed.

4.2 Observation Table Template

The following table can be filled in during the practical demonstration. Record the approximate object distance, the Serial Monitor reading, the lid response and any observations.

Test	Object position	Expected response	Actual observation
1	About 10 cm	Lid opens	To be recorded
2	About 20 cm	Lid opens if measured ≤ 20 cm	To be recorded
3	More than 20 cm	Lid closes	To be recorded
4	No object / no echo	Check safe response and reading	To be recorded

4.3 Advantages

- Touch-free operation: the user does not need to lift the lid directly.
- Convenience: the lid can open when a hand or object approaches the sensor.
- Educational value: the prototype demonstrates sensor interfacing, embedded programming and actuator control.
- Simple control logic: the distance threshold and servo angles are easy to understand and modify.
- Accessible components: the design uses common modules and connecting wires.
- Expandable design: additional sensors or alerts can be explored as future improvements.

4.4 Limitations

The sensor estimates distance and does not identify whether the object is a hand, waste item or another object. The lid can therefore open whenever an object enters the detection range. Ultrasonic readings may be affected by the angle and surface of the object or by the sensor's position. The servo may not provide enough torque for a heavy lid, and the mechanism may need adjustment. The present design does not detect when the bin is full, send notifications, separate waste or log usage.

4.5 Future Scope

A second ultrasonic sensor could be installed inside the bin to estimate the fill level. A buzzer or indicator could warn the user when the bin approaches capacity. A suitable wireless module could be added to send a full-bin notification to a phone or dashboard. Other improvements include a more robust enclosure, improved power management, protection against moisture, a stronger actuator for a heavier lid and a more reliable response to invalid sensor readings.

4.6 Environmental and Practical Relevance

A smart bin can support awareness of convenient and hygienic waste disposal, but automation alone does not solve waste collection or segregation. The prototype should be used with appropriate waste-handling practices, and its electronic parts should be kept away from wet or corrosive waste. Batteries and electronic components should be disposed of responsibly when they reach the end of their service life.

4.7 Further Development Plan

- Improve the lid linkage and test repeated open-close cycles.
- Add an echo timeout and explicit handling of invalid sensor readings.
- Test the system at multiple distances and record actual observations.
- Add a fill-level sensor and alert only after the basic lid mechanism works reliably.
- Evaluate a suitable power arrangement for the Arduino and servo under real operating load.

CONCLUSION AND REFERENCES

CONCLUSION

The Smart Dustbin project demonstrates the use of an Arduino Uno, HC-SR04 ultrasonic sensor and servo motor to automate the opening and closing of a dustbin lid. The sensor measures the approximate distance to an approaching object, the Arduino compares the measurement with the 20 cm threshold and the servo responds by moving to the programmed open or closed position. The Serial Monitor provides distance readings that help the builder observe the operation of the sensor.

The project combines hardware assembly, programming, distance calculation and mechanical movement in a simple educational prototype. It can reduce the need for direct contact with the lid and provides a practical introduction to automation. Its dependable operation requires suitable wiring, stable power, accurate sensor placement and a freely moving lid. Future enhancements such as fill-level monitoring, audible alerts, wireless notifications and improved error handling could increase the usefulness of the system.

REFERENCES

1. Arduino. Arduino Uno hardware documentation.

2. Arduino. Servo library reference and examples.

3. HC-SR04 ultrasonic ranging module technical documentation.

4. Smart Dustbin hardware component list and Arduino sketch supplied for this project.